



Introduction

1.1 Problem Definition

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1.1 Problem Definition

In traditional document search processes, users must know the specific file name and manually open and search each PDF file to locate relevant content. This becomes increasingly cumbersome and time-consuming as the number of files grows, especially in large organizations with vast document repositories like company policies, contracts, or technical manuals.

To address this, the current project seeks to develop an AI-powered PDF Content Extraction and Retrieval System. This application allows users to simply input a query, and the system automatically searches across multiple PDF documents, retrieves relevant content based on keyword matches and AI-driven similarity scores.

1.2 Objective of the System

The objective of this system is to streamline the process of searching and retrieving relevant information from a large collection of PDF documents. By leveraging advanced AI techniques, this system aims to simplify the task of content extraction, enabling users to efficiently locate information without needing prior knowledge of specific file names or manually scanning documents. This approach will enhance productivity and reduce the time spent on document analysis.

1.3 PROJECT PROFILE

Project Name: Techforce Buddy

Type of Application: Web Application Project

Project Description:

In organizations, policy documents are often lengthy and contain complex structures with headers, footers, and repetitive sections, making it challenging for employees to quickly find specific information.

To address this, I will use an Automated **PDF Content Extraction**. By automating content extraction from PDF files and implementing a keyword-based search system, this project provides users with the ability to retrieve relevant sections or paragraphs quickly and accurately.

This project focuses on building a robust system for extracting relevant content from PDF documents. The system automates the extraction of paragraphs or sections based on a set of keywords provided in user queries, offering a solution for efficient policy search and retrieval.

The challenge lies in dealing with large volumes of PDF documents that contain structured and unstructured text, including repetitive headers, footers, and non-relevant sections. This system uses advanced text extraction techniques to remove irrelevant information such as headers and footers.

To address this, the project integrates Natural Language Processing (NLP) techniques and leverages tools like Apache PDFBox for text extraction and OpenNLP for text preprocessing. The goal is to ensure that users can quickly retrieve precise and meaningful policy information by querying with relevant keywords, eliminating the need to manually sift through extensive documents.

Team Size: 1

Mentor: Mr. Manish Mori

1.4 ADVANTAGES AND LIMITATIONS OF THE SYSTEM

Advantages

Efficient Search Across Multiple Files: The system allows users to search across multiple PDF files at once, significantly reducing the time required to find specific information.

Simplified User Interaction: Users can retrieve relevant content without knowing the exact file name or location, enhancing usability for non-technical users.

Automation of Tedious Tasks: Automating content extraction and analysis helps users avoid manually scanning large PDFs or documents, improving workflow efficiency.

Optimized Assets Utilization: Asset management systems help organizations maximize asset utilization by identifying underutilized assets, scheduling maintenance proactively, and reallocating resources based on usage patterns.

Disadvantages

Structured Data Organization: Organize PDF files into a structural format for easier information retrieval.

Complexity and Training: Asset management systems may have a steep learning curve for users, requiring training and change management efforts to ensure successful adoption across the organization.

Maintenance and Support: Ongoing maintenance, updates, and technical support for the asset management system are necessary to address issues, implement enhancements, and ensure system reliability. This requires allocation of resources and ongoing investment.

Dependency on technology: Organizations become increasingly reliant on technology for asset management, which can pose risks such as system outages, data loss, cybersecurity threats, and software obsolescence.

2

Requirement Determination & Analysis

2.1 Requirement and Determination

2.2 Targeted User

2.1. REQUIREMENT DETERMINATION

Features:

Admin can:

Login

Pre-process data

Train Model

Search the query

End User can:

Login

Register

Search the query

TOOLS AND TECHNOLOGIES:

Front End: HTML, CSS, BOOTSTRAP, Thymeleaf, Javascript

Back End: JAVA (21)

Framework:SpringBoot

Database: MySQL

AI libraries: OpenNLP, Deeplearning4j

2.2 TARGETED USERS

Administrators: Responsible for uploading pdf files, preprocessing the data of pdfs and training the model using data which is preprocessed.

End User: End users are typically non-technical users such as employees or HR personnel who need to quickly retrieve relevant policy or document information without having to know the file name or manually search through multiple documents. End users input search queries, and the system returns the most relevant content from the documents based on the query.

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System Design

3.1 Use Case Diagram

3.2 Activity Diagram

3.3 Entity Relationship Diagram

3.4 Data Dictionary

3.1 USE CASE DIAGRAM

A use case diagram is a graphical representation of the interactions between actors (users or external systems) and a system. It illustrates the various use cases (functionalities or tasks) that the system performs to achieve specific goals for its users. Use case diagrams are typically used during the early stages of software development to capture and communicate the requirements of a system in a clear and understandable manner.

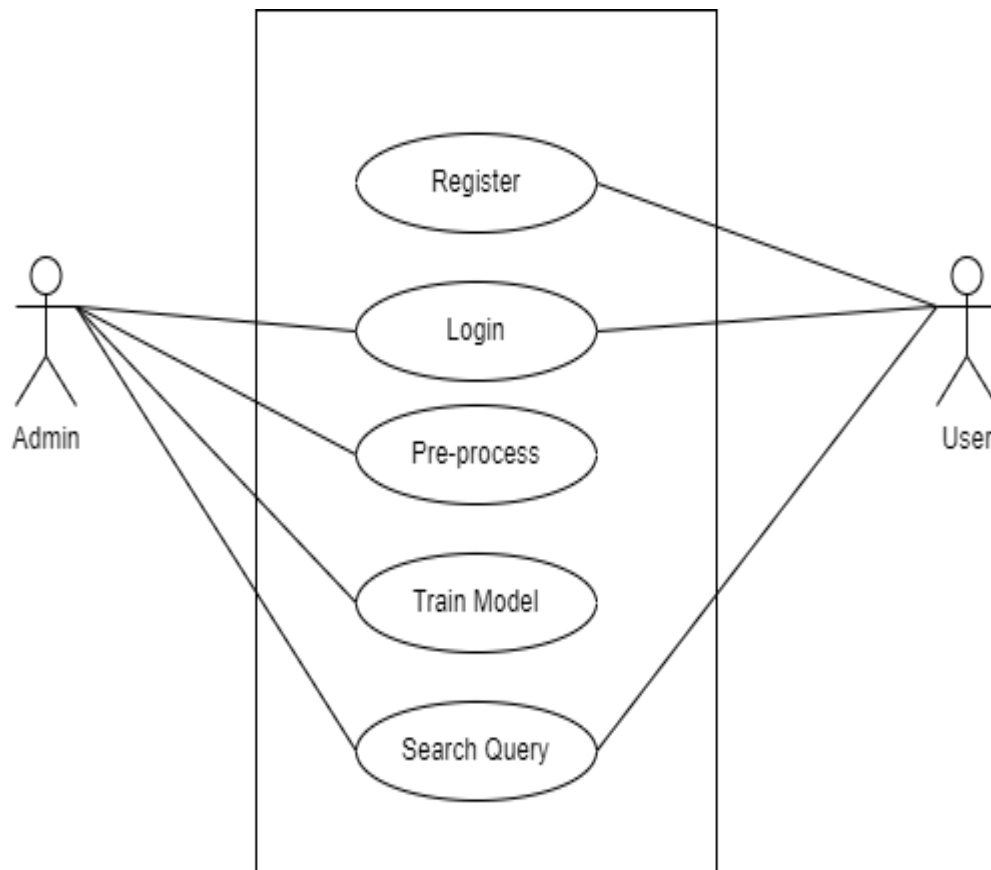


Figure 3.1. Use Case Diagram

3.2 ACTIVITY DIAGRAM

It is the Equivalent to flow chart diagram. It showing flow of control from activity to activity. It is used for modeling the logic captured by a single use-case Scenario.

Activity diagram is used for simple and perspective Illustration of what happens in a work flow, what activities can be done in parallel & whether there is alternative path through the workflow. It shows the workflow from the start point to the finish point.

Login Activity

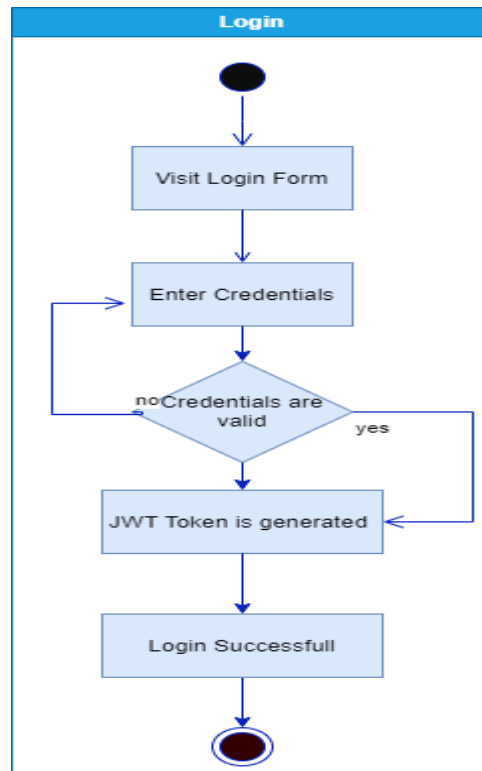


Figure 3.2.1 Login Activity

Register Activity

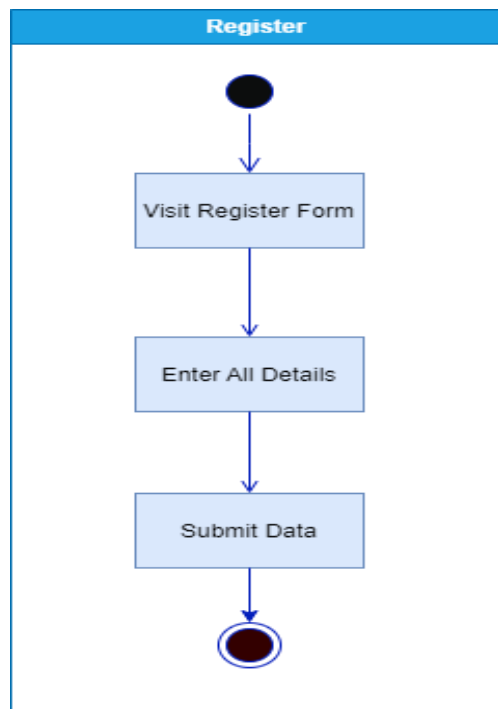


Figure 3.2.2 Register Activity

Pre-process data Activity

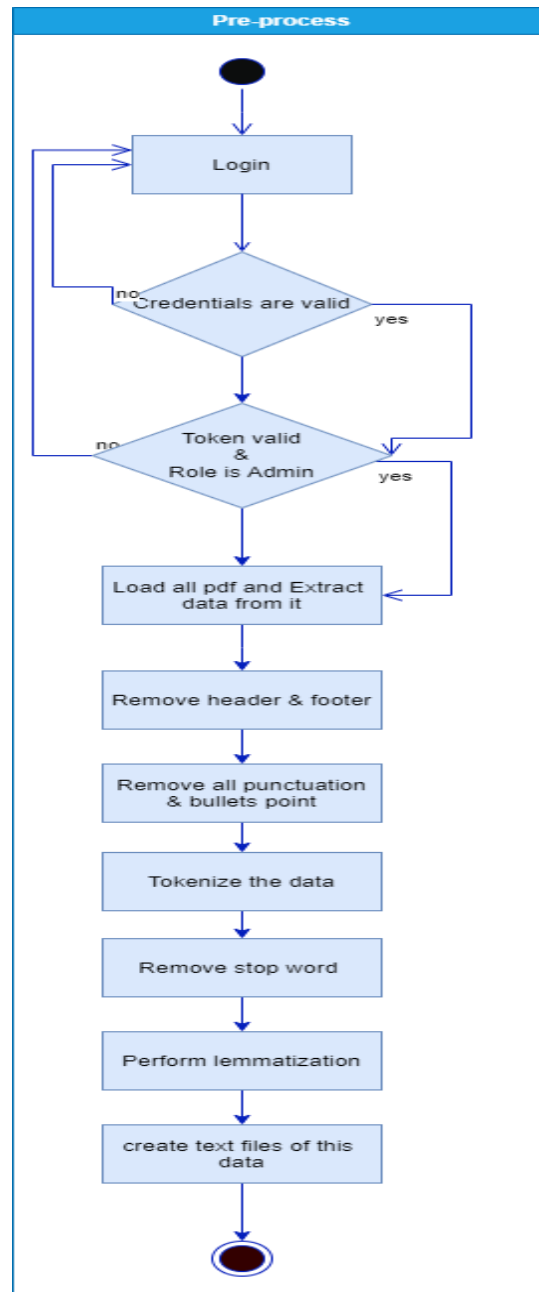


Figure 3.2.3 Pre-process data activity

Train model Activity

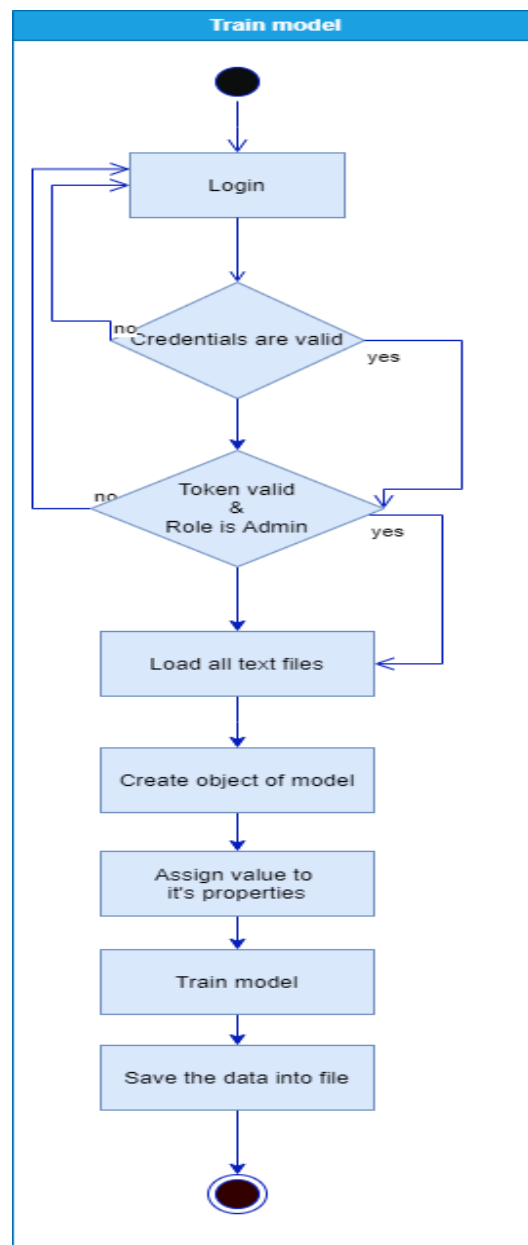


Figure 3.2.4 Train model activity

3.3 ENTITY RELATIONSHIP DIAGRAM

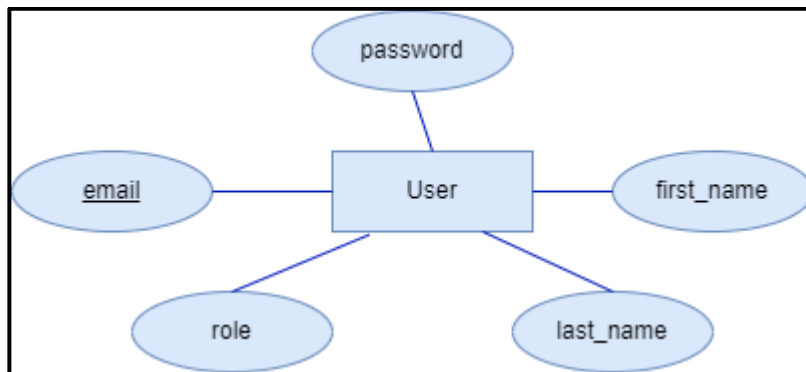


Figure 3.3 E-R Diagram

3.4 DATA DICTIONARY

User Table

Table 3.4.1 User Table

Filed type	Data type	Size	Constraints	Description
email	varchar	150	Not null	Email id of user
first_name	varchar	50	Not null	First name of user
last_name	varchar	50	Not null	Last name of user
password	varchar	255	Not null	Password for login
role	varchar	255	Not null	Role of the user

4 Development

4.1 Coding Standards

4.2 Working Project Screenshot

4.1 CODING STANDARD

Folder Structure

Describe the organization of files and folders in the project.

Naming Conventions:

Specify the naming conventions for packages, classes and interfaces, variables, methods, constants, annotations etc.

Documentation Format

Explain the format used for documenting code, such as JavaDoc.

Indentation and Formatting

Use consistent indentation and formatting to make your code

Error Handling

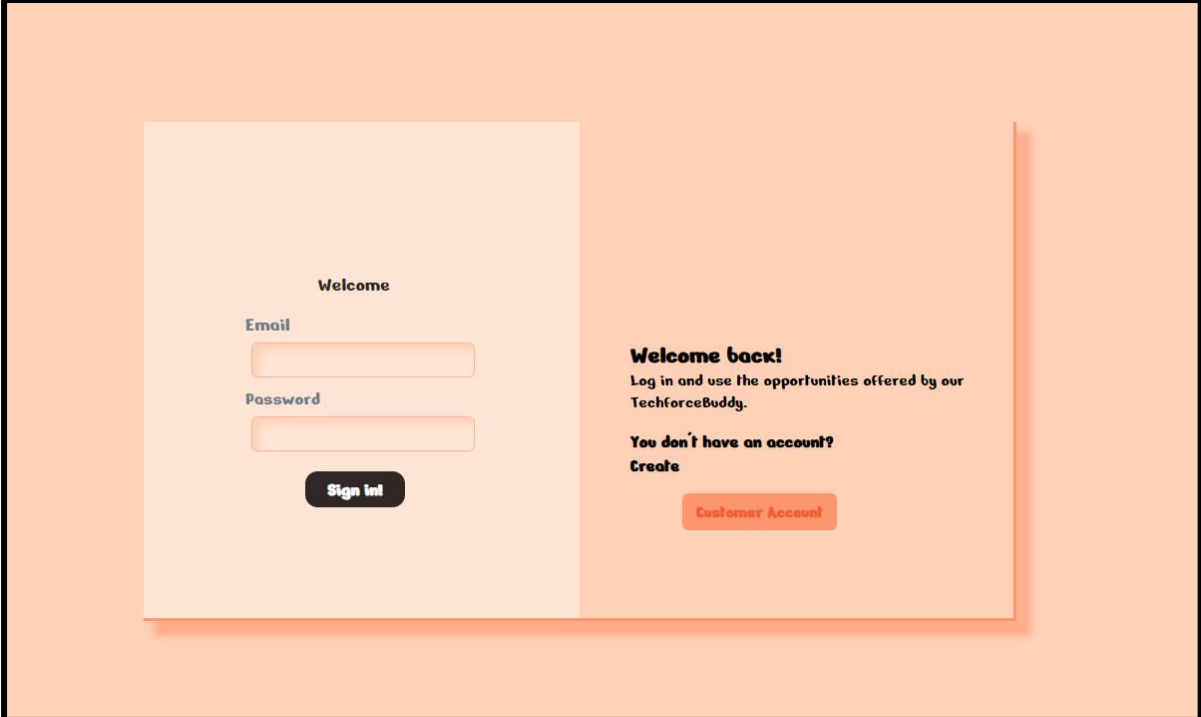
Describe the overarching strategy for error handling in the project, including Prevention: Techniques and practices used to prevent errors from occurring in the first place. Detection: Methods for detecting errors during runtime.

Comments

Example: Inline comments explaining complex logic.

4.1 WORKING PROJECT SCREENSHOTS

Login

A screenshot of a login page with an orange background. On the left, a white card contains the text "Welcome" above two input fields labeled "Email" and "Password". Below these fields is a dark "Sign in!" button. On the right, the text "Welcome back!" is followed by "Log in and use the opportunities offered by our TechforceBuddy." Below this, it says "You don't have an account?" followed by a "Create" link and a "Customer Account" button.

Welcome

Email

Password

Sign in!

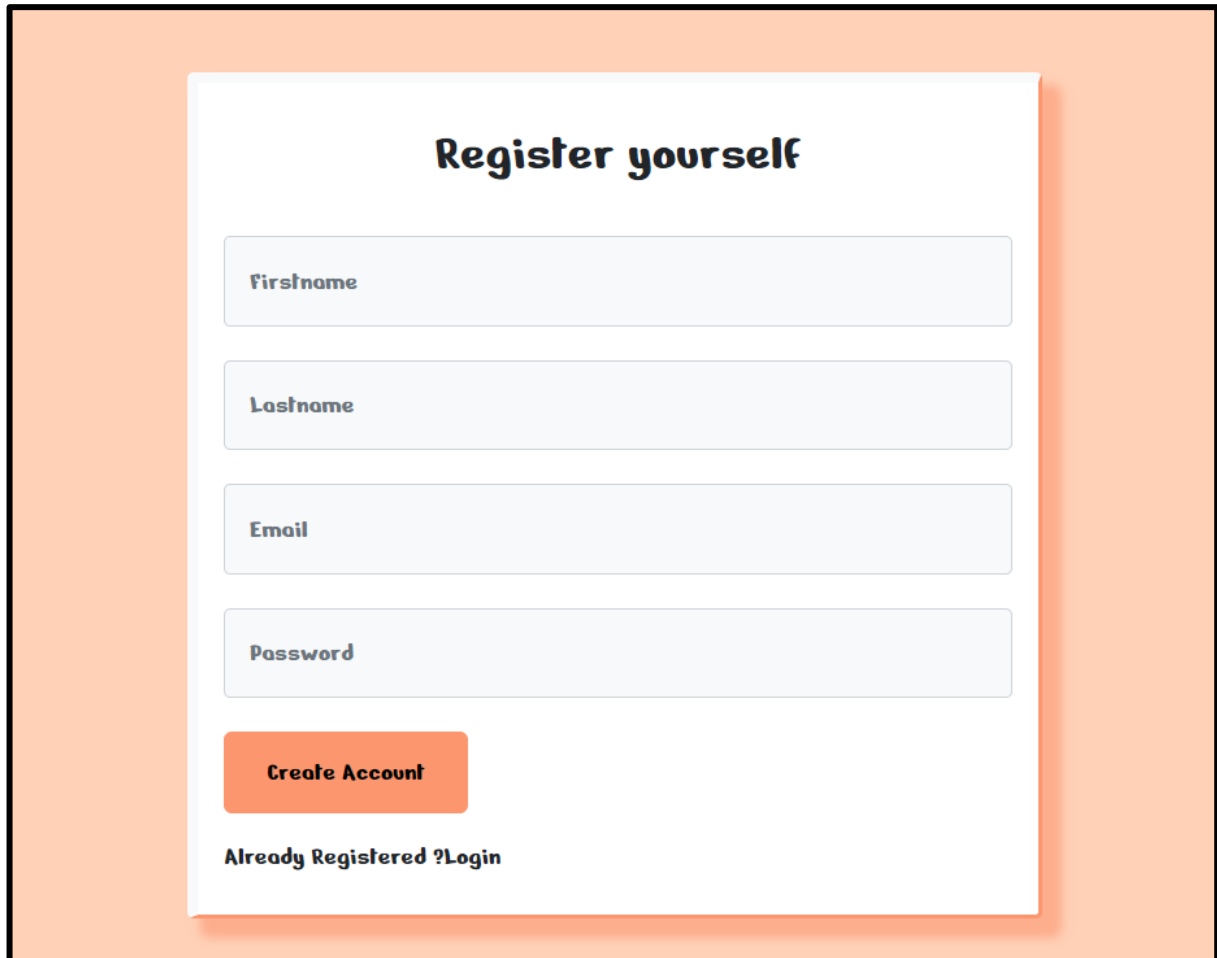
Welcome back!
Log in and use the opportunities offered by our TechforceBuddy.

You don't have an account?
[Create](#)

[Customer Account](#)

Figure 4.1.1 Login Page

Registration Page



Register yourself

Firstname

Lastname

Email

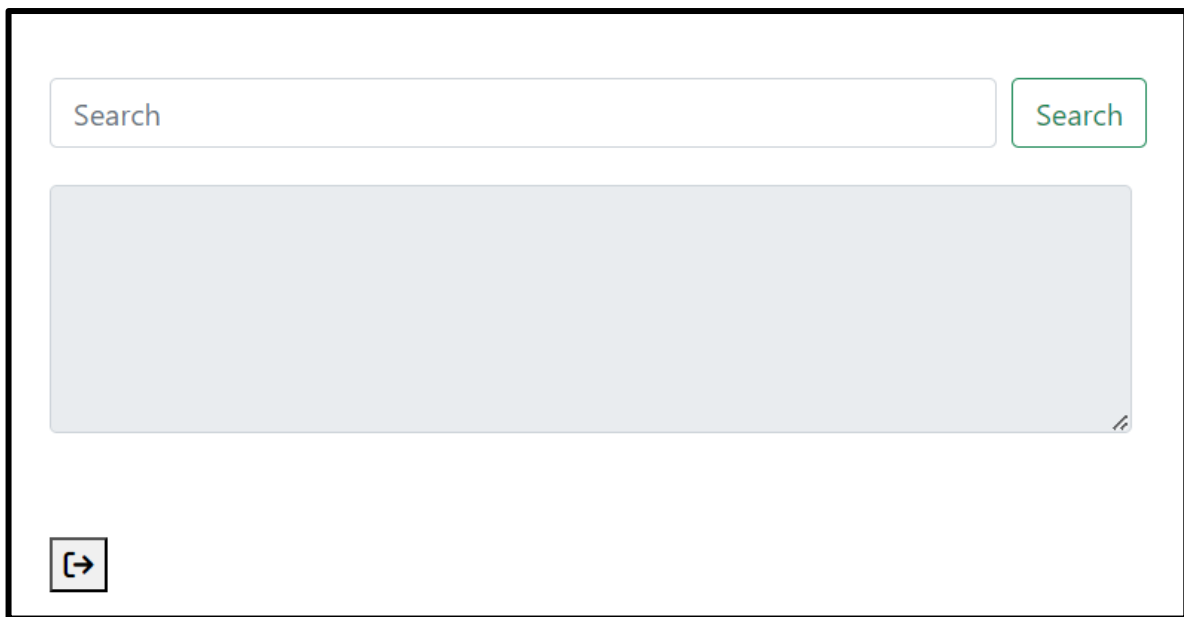
Password

Create Account

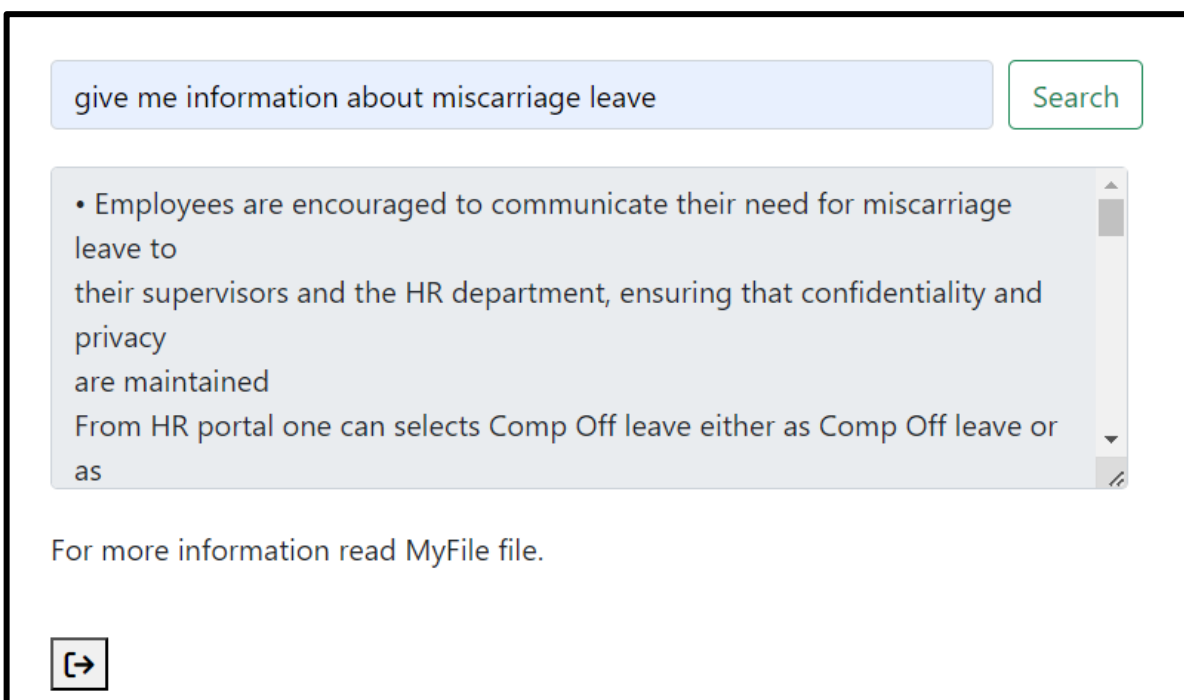
Already Registered ?Login

Figure 4.1.2 Registration Page

Search Page



This image shows an empty search interface. It features a text input field with the placeholder text "Search" and a green "Search" button to its right. Below the input field is a large, empty light gray rectangular area. At the bottom left of the interface is a small square button containing a right-pointing arrow inside a square frame.



This image shows the search interface with a query and results. The text input field contains the query "give me information about miscarriage leave". The results are displayed in a light gray box with a vertical scrollbar on the right. The results include a bulleted list item and a paragraph. Below the results box is a line of text. At the bottom left is the same square button with a right-pointing arrow.

give me information about miscarriage leave

Search

- Employees are encouraged to communicate their need for miscarriage leave to their supervisors and the HR department, ensuring that confidentiality and privacy are maintained

From HR portal one can select Comp Off leave either as Comp Off leave or as

For more information read MyFile file.

Figure 4.1.3 Search Page

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Proposed Enhancement

- To further enhance the system's capabilities, future developments will focus on:

1. Structured Data Organization

- ✓ Organize PDF files into a structural format for easier information retrieval
- ✓ Utilize techniques such as named entity recognition and entity disambiguation

2. Advanced Information Retrieval

- ✓ Implement TF-IDF or Lucene algorithms for faster and more accurate information retrieval
- ✓ Explore other information retrieval techniques, such as elastic search

3. Sentiment Analysis and Context Understanding

- ✓ Integrate Natural Language Processing (NLP) for sentiment analysis and context understanding
- ✓ Analyze user queries to detect emotions, tone, and intent

4. Context-Aware Response Generation

- ✓ Utilize NLP to rewrite responses based on context and sentiment
- ✓ Generate more personalized and empathetic responses to user queries

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Conclusion

The **Automated PDF Content Extraction and Keyword-Based Retrieval System** provides an efficient solution to the challenges of retrieving relevant information from large collections of PDF documents. By leveraging advanced techniques such as natural language processing and machine learning (Word2Vec), the system allows users to search and find relevant sections without needing to know the specific file names or manually read through extensive documents. This significantly reduces the time and effort required to locate important information, making the process streamlined and user-friendly.

By addressing the limitations of manual searching, the system offers substantial benefits to both technical and non-technical users. However, challenges such as processing complex document structures and ensuring accurate similarity measures remain areas for further improvement. Overall, this project demonstrates a practical and impactful application of AI-driven search technologies in document management and retrieval systems.

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